

EEE 435 / EEE 4117 (Project Description Involving Complex Engineering Problem)

1. a) Design a 2-stage OPAMP (a differential amplifier followed by a common-source amplifier) with the following specifications. Use cadence tool and TSMC 45nm pdk for the design purpose. Also, use 1.2V transistors from the pdk library.
 - i) Voltage gain = 55 to 70 dB
 - ii) Phase margin = 60° to 70°
 - iii) Bandwidth = 50 to 60 MHz
 - iv) ICMR- : 0.3 to 0.6V, ICMR+: 0.9 to 1.0V
 - v) Output load capacitance, $C_L = 1$ to 5pF
 - vi) Slew rate, SR = 5V/ μ s to 20V/ μ s

b) While designing the OPAMP bias transistors from a supply voltage of 1.2V and a single current source. Minimize DC power consumption as much as possible.

c) Also, perform DC simulation, AC analysis with sweeping frequency from 1 to 10 GHz of frequency, transient analysis and noise analysis.

d) Calculate the output impedance and gain of each stage using parameters obtained from DC simulation and compare the calculated results with the simulation values.
2. Prepare a report which shows the
 - i) calculation of aspect ratio of all transistors 7
 - ii) snapshot of the circuit, showing the operating region, bias current and g_m of all transistors (from DC simulation) 3
 - iii) AC analysis with gain and phase plot. Identify the unity gain frequency and phase margin. 4
 - iv) Transient analysis: apply a sinusoidal signal of 5 to 10mV (peak-to-peak) on top of the DC common mode voltage at the input. Plot the input and output voltages. 3
 - v) Noise analysis: perform noise analysis with sweeping frequency of 1 to 10GHz. Plot output and input referred noise. 3